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<Application of Machine Learning Models to Mortality Data Analysis During the COVID-19 Pandemic in Cyprus>

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[ATHENS, APRIL/2025]

Acknowledgements

I would like to thank my professors, Petros Stefaneas and Haris Dakolias, for their assistance and guidance throughout my project. Their support made the process clearer and more manageable.

Abstract

Predicting COVID-19 mortality is a critical challenge for public health, as the virus has caused millions of deaths worldwide. Numerous factors such as age, underlying health conditions, obesity, and the quality of medical care have been identified as key determinants of disease outcomes. While Artificial Intelligence has been widely used in analyzing epidemiological data, understanding how specific variables affect mortality prognosis remains an open research field.

This study investigates the prediction of COVID-19 mortality using Machine Learning methods. To improve model performance, feature selection was conducted using the Sequential Forward Floating Selection (SFFS) method combined with 5-fold cross-validation (CV), in order to identify the most influential variables affecting prediction. Subsequently, various classifiers (MLP, Random Forest, SVM, LG) were trained, and statistical tests (Wilcoxon, Friedman, Mann-Whitney U) were conducted to compare models and evaluate the significance of features. The MLP emerged as the optimal model, achieving nearly the same performance as the Random Forest (ROC-AUC = 0.9878 vs. 0.9903), while using fewer features. The optimal hyperparameter tuning of the MLP model led to high values in accuracy, F1 score, recall, AUC-ROC, and precision.

The study's findings demonstrate that the combined use of SFFS and 5-fold CV can contribute to the development of more efficient and interpretable models for predicting COVID-19 mortality.

Keywords: Mortality prediction, COVID-19, Machine Learning, Artificial Intelligence, Classifiers, Feature selection, Statistical tests, F1 score, ROC-AUC, Accuracy, Recall, Precision.

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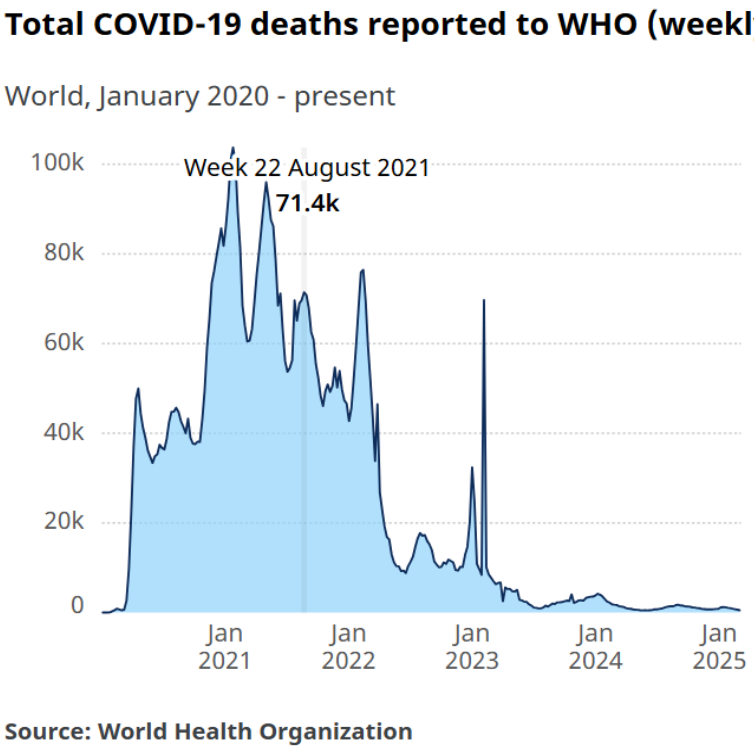
Introduction

COVID-19

The COVID-19 disease was caused by the SARS-CoV-2 virus and first emerged in Wuhan, China, in late 2019. It was reported to the World Health Organization (WHO) on December 31, 2019. Since then, it has spread globally and evolved into a pandemic that is ongoing to this day. COVID-19 is primarily transmitted through close contact with infected individuals. Diagnosis of the virus is carried out either through PCR tests for molecular detection or via self and rapid antigen tests. Vaccination remains the primary preventive measure.
According to data from the World Health Organization, the total number of recorded deaths is 7,089,989. [2]



The Role of Artificial Intelligence in Predicting COVID-19 Deaths

Artificial Intelligence has proven to be extremely useful in predicting deaths among hospitalized COVID-19 patients. Machine Learning, as a subfield of AI, enables the analysis of clinical patient data to predict the likelihood of death. By identifying the variables that influence mortality risk, machine learning helps improve the preparedness of healthcare systems and the more efficient allocation of resources.

Objective

In this study, we train various machine learning models using a dataset of individuals infected with COVID-19 in Cyprus during the 2020–2025 period, aiming to predict which patients are at greater risk of death. We select the optimal model with the highest accuracy and conduct statistical tests to evaluate its performance.

Chapter 1
Machine Learning

Machine Learning is a branch of Computer Science and more specifically falls under the field of Artificial Intelligence. The term was defined in 1959 by Arthur Samuel as “the field of study that gives computers the ability to learn without being explicitly programmed.” Machine Learning focuses on the study and development of algorithms that can learn from data and make predictions based on it. [3]

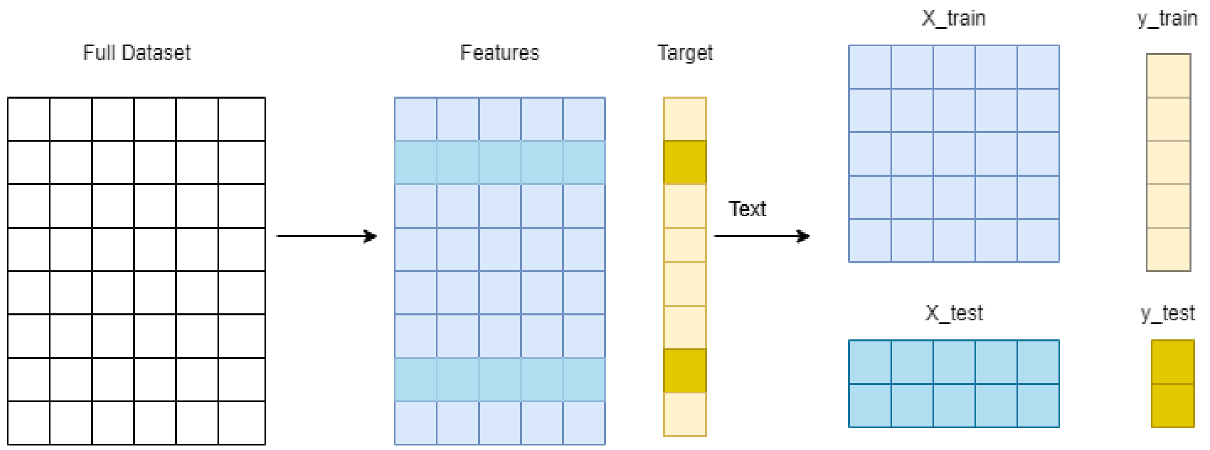
In contrast to classical programming, where rules are explicitly defined by the programmer, in Machine Learning the system learns from data and dynamically adapts. Classical programming relies on the creation of algorithms with predefined logic, a process that can become complex and difficult to scale when new inputs arise. On the other hand, Machine Learning uses data as the foundation to train models, allowing for automatic adaptation and generalization of results. Thus, while classical programming is ideal for problems with clearly defined rules, Machine Learning is more effective when the rules are complex or unknown.

Supervised Learning

Supervised learning is the process by which an algorithm constructs a function that maps given inputs to known desired outputs, with the ultimate goal of generalizing this function to inputs with unknown outputs. [4]

Training a supervised learning model follows the steps below:

1. Data splitting: The dataset is divided into a training set and a test set, usually in a ratio of 70–80% for training and 30–20% for testing.
2. Model training: The model receives the training set samples as input, along with their corresponding labels, and gradually learns the relationships between features and outputs.
3. Model evaluation: The test set samples are fed into the model without labels, and the model’s output is compared to the actual values.
4. Performance assessment: The model’s output is compared against the actual test set labels, and its performance is evaluated using statistical metrics.



This procedure is applied to both classification and regression problems. In classification, the model output belongs to discrete categories, whereas in regression the output is a continuous numerical value.

Supervised Learning Algorithms:

Logistic Regression:

Logistic Regression is a supervised learning algorithm used for solving binary classification problems. The model is based on the logistic function [5]:

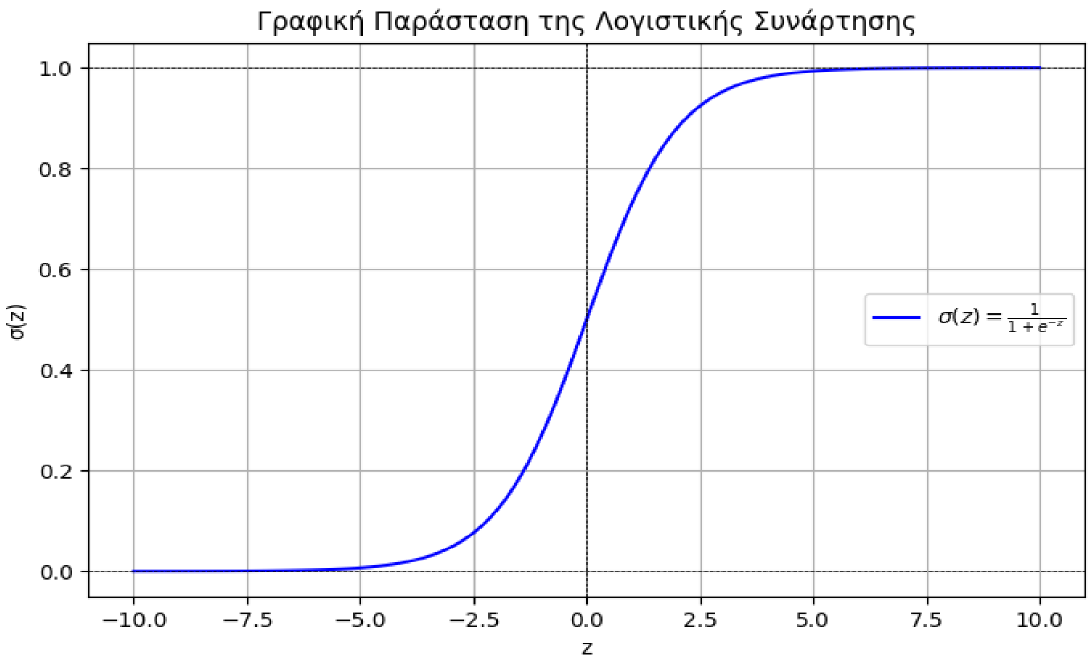
$$\sigma(z) = \frac{1}{1+e^{-z}}$$

When $z=w^T x+b$. In the above equation:

- w represents the weight vector of the model
- x represents the feature vector, $x \in \mathrm{R}^n$, (where n is the number of features)
- b is a constant

Regardless of the value of x, the output of the logistic function is constrained to the interval [0, 1] and represents the probability of the observation belonging to class 1.

The following graph illustrates the logistic function σ(z) , for $z \in (-10.0, 10.0)$



Let us assume that we have a dataset{ (x_i , y_i) } ,where:

- x_i represents the features of the observation.
- y_i represents the true label (either 0 or 1)

In logistic regression, we attempt to classify each sample x_i into one of two categories (0 or 1).

The probability that a sample x_i belongs to category 1 is given by:

$$P(y_i = 1 \mid x_i) = p = \sigma(w^T x + b) = \frac{1}{1+e^{-w^T x+b}}$$

While the probability of belonging to category 0 is:

$$P(y_i = 0 \mid x_i) = 1 - p$$

The probability that each sample is classified correctly is as follows:

$$P(x_i;w,b) = p^{y_i} (1-p)^{1-y_i}$$

The total probability for all observations is the product of the probabilities of all samples:

$$\prod_{i=0}^{N} P(x_i \; ; w, b)$$

This quantity expresses the probability that all observations are correctly classified. The goal is to estimate the parameters w and b that maximize this probability, so that the model produces the highest possible probabilities for the correct labels.

However, the product is difficult to optimize directly, so we take the natural logarithm (log-likelihood):

$$L(w, b) = \sum_{i=0}^{N} \left(y_i \log(p) + (1-y_i)\log(1-p) \right)$$

To convert the problem into a minimization one, we take the negative log:

$$L(w, b) = -\sum_{i=0}^{N} \left(y_i \log(p) + (1-y_i)\log(1-p) \right)$$

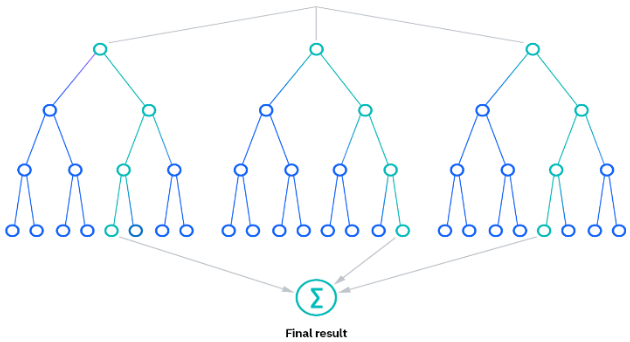
The quantity obtained from this process is called the **Binary Cross-Entropy Loss**, and minimizing it can be achieved through various optimization algorithms, such as:

- Gradient descent
- L-BFGS
- Stochastistic average gradient
- SAGA

The final prediction is based on the value of $\sigma(w^T x + b)$. If it is greater than 0.5, the model predicts that the observation belongs to category 1. Otherwise, it classifies it into category 0.

Random Forest

Random Forest is an algorithm used for both classification and regression problems. It belongs to the category of ensemble algorithms, as it does not rely on a single decision tree, but on a collection of trees (forest) that work together to improve the prediction accuracy and reduce the risk of overfitting. [6]



The Random Forest training algorithm applies the general technique of bootstrap aggregation (bagging) in order to improve prediction accuracy and reduce sensitivity to noise.

Given a training set X={ $x_1,\ldots,x_n$ } with corresponding responses Y={ $y_1,\ldots,y_n$} , a random subset of data is selected from the original training set (X,Y) allowing for replacement. This new subset of data is denoted as { X_b,Y_b }. The bagging method repeats this process B times, where B is determined by the user through the hyperparameter n_estimators.

In classification problems, Random Forest selects the category with the most votes, while for regression problems, it returns the average of the predictions.

In the classical bagging method, decision trees often tend to be correlated because if there are very strong features, they will be almost always selected by most of the trees. To reduce this correlation, Random Forest uses a modified training process, where at each node of the tree, instead of examining all available features, a random subset of features is selected. This process is called "feature bagging" and is controlled by the hyperparameter max_features\text{max_features}max_features, which determines how many features will be used at each split.

Random Forest is particularly suitable for large datasets, as it combines the ability to handle large amounts of data with efficiency in prediction, while reducing overfitting. [7]

Support Vector Machines (SVM)

Support Vector Machine (SVM) is a machine learning algorithm primarily used for classification, but it can also be applied to regression problems. The goal of the Support Vector Machine algorithm is to find an optimal hyperplane that separates different classes of data points in a high-dimensional space. It is particularly effective in solving binary classification problems, where the objective is to classify data points into one of two categories. [8]

Mathematical Formulation:

Let’s assume we have a training dataset consisting of N samples:

$$(x_i , y_i) , i= 1, 2, \dots, N$$

Where:

- x_i represents the feature vector of the i-th data point,
- $y_i \in \{-1,1\}$ is the label of the i-th data point, with the goal of determining the optimal hyperplane that separates the two classes. This hyperplane is defined by the equation:

$$w^T x + b = 0$$

Where:

- w is the weight vector
- b ε is the bias term (intercept)

Binary Classification with Support Vector Machines (SVM)

In binary classification, SVM classifies data points based on the sign of the following equation:

$$g(x) = w^T x + b$$

- If $w^T x + b \geq 1$, then the sample belongs to class **+1**
- If $w^T x + b \leq -1$, then the sample belongs to class **-1**

The support vectors are the data points that lie on the boundaries of classification, i.e., on the hyperplanes:

$$w^T x + b = +1, \quad w^T x + b = -1$$

The distance between these two hyperplanes is called the margin, and it is given by:

$$\frac{2}{||w||}$$

The optimal separating hyperplane is the one that maximizes this margin, ensuring the best possible separation between the two classes.

Optimization – Maximum Margin

The task of finding the hyperplane that maximizes the margin is formulated as the following minimization problem:

$$\min \frac{1}{2}||w||^2$$

subject to the constraint:

$$y_i (w^T xi + b) \geq 1 \ , \ \forall\, i$$

This constraint ensures that all data points lie either on or outside the classification boundaries.

We define the Lagrangian cost function as:

$$L(w, b, \lambda_1 ,\ldots, \lambda_p) = \frac{1}{2}||w||^2 - \sum_{i=1}^{P}(\lambda_i [y_i (w^T x_i + b) – 1])$$

$$\lambda_i \geq 0 \ , \ i=1,\ldots,P$$

This function must be minimized with respect to w and b, and maximized with respect to λ_i.

Karush-Kuhn-Tucker (KKT) Conditions:

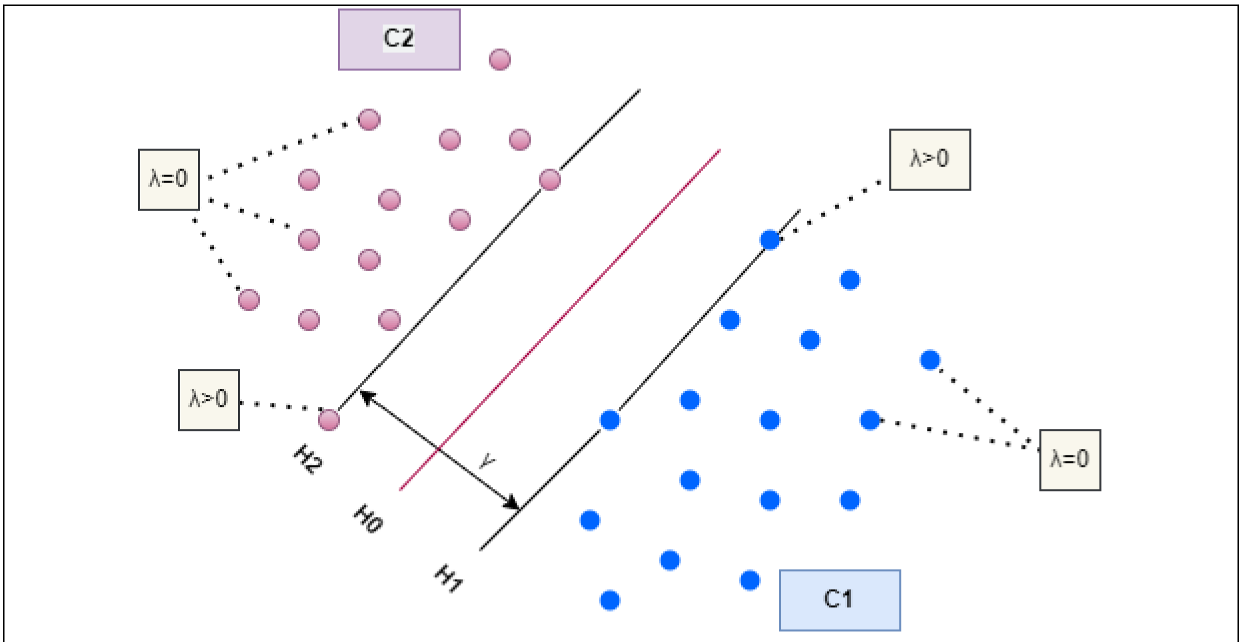
- $\frac{\partial \mathrm{L}}{\partial \mathrm{b}} = 0 \ \rightarrow \ \sum_{i=1}^{P}(\lambda_i y_i) =0$
- $\frac{\partial \mathrm{L}}{\partial \mathrm{w}} = \ \rightarrow \ w = \sum_{i=1}^{P}(\lambda_i y_i x_i)$
- $\lambda_i [y_i (w^T x_i + b) – 1] \geq 1$

Optimal Separating Function

The optimal separating function is given by:

$$g^*(x) = w^T x +b = \sum_{i=1}^{P}(\lambda_i y_i x_i^T)x + b$$

Only those multipliers λ_i that correspond to support vectors xi can be non-zero. For all other points, λ_i=0.



H0 : Hyperplane

H1,H2 : Support Vectors

C1,C2 : Classes

$\gamma = \frac{2}{||w||}$

Soft Margin SVM

In practice, data is not always perfectly separable. To allow for some classification errors, slack variables ξ_i , are introduced, permitting some samples to violate the margin. The new optimization problem is formulated as:

$$y_i (w^T x_i + b) \geq 1 - \xi_i, \quad \forall i$$

Where:

- C is a hyperparameter that controls the trade-off between maximizing the margin and minimizing classification errors.
- If $\xi_i=0$, the sample is correctly classified and outside the margin.
- If $0<\xi_i\leq 1$, the sample lies within the margin but is still correctly classified.
- If $\xi_i>1$, the sample is misclassified.

The value of C significantly affects the behavior of the classifier:

- Large C: Emphasizes correct classification, but may lead to overfitting.
- Small C: Allows more classification errors, which may improve model generalization.

Kernel Trick in Support Vector Machines

Support Vector Machines (SVMs) rely on the principle of identifying a hyperplane that separates data into different classes effectively. However, when the data is not linearly separable, the Kernel Trick is used.

This technique applies non-linear transformations to the original feature space, projecting the data into a higher-dimensional space where it becomes linearly separable. Using this approach, SVMs can construct non-linear, curved decision boundaries that more accurately capture the underlying structure of the data, thus reducing classification errors.

Among the most commonly used kernels are:

- Radial Basis Function (RBF) Kernel: Computes the similarity between data points based on their distance from one or more central points, generating new features that help distinguish between classes.
- Polynomial Kernel: Constructs polynomial combinations of the input features, allowing the model to capture more complex patterns in the data.

Multi-Layer Perceptron (MLP)

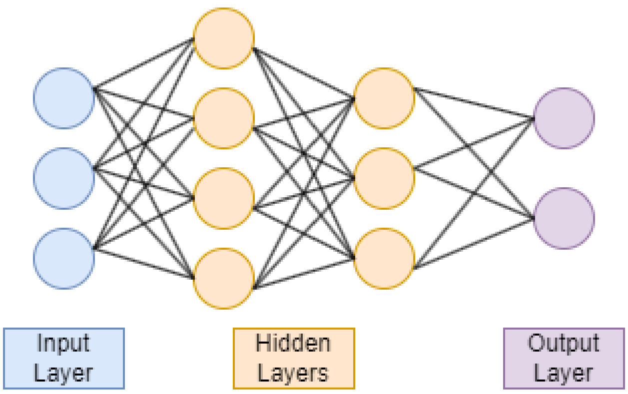
The Multi-Layer Perceptron (MLP) belongs to the category of Artificial Neural Networks (ANNs) and is one of the most widely used types of neural networks. MLP is based on a supervised learning process aimed at creating a non-linear prediction model. [9]

Structure of MLP

The MLP consists of three main parts:

- Input Layer
- Hidden Layers
- Output Layer

All layers are fully connected, meaning that each neuron in one layer is connected to every neuron in the next layer.



Computational Flow of MLP

Assume the MLP has P input-target pairs:

$$(x^{(p)}, t^{(p)}), \quad p=1, \ldots, P$$

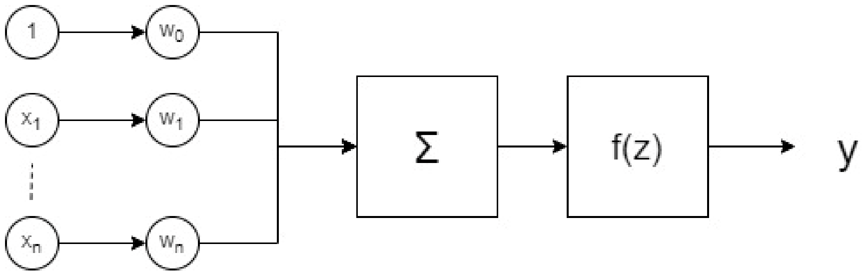
Where $x^{(p)}$ is the input vector, $t^{(p)}$ is the target vector, and $y^{(p)}$ is the output vector.

The computation process includes the following steps:

1. Each input $x_i^{(p)}$ is multiplied by the corresponding weight wi, and the total weighted sum is calculated:

$$z = \sum_i w_i x_i + w_0$$

2. A non-linear activation function f(z) is applied to the result.



3. The output of each layer is used as input for the next layer.
4. The final result from the output layer represents the predicted value of the model.

Activation Functions

The most commonly used activation functions include:

- Rectified Linear Unit (ReLU) : f(z) = max (0,z)
- Sigmoid : f(z) = $\frac{1}{1+e^{-z}}$
- Unit Step Function (Heaviside Step Function) : f(z) = $\begin{cases} 0, & z < 0 \\ 1, & z \geq 0 \end{cases}$

Training the MLP

The model's performance is evaluated using a **cost function**, which measures the error between the actual and predicted outputs:

$$J_{MLP} = \frac{1}{P}\sum_{p=1}^{P} || t^{(p)} - y^{(p)} ||^2$$

The goal of training is to minimize the cost function, which is achieved using the Backpropagation algorithm. This algorithm computes the output error and propagates it backward through the network, leading to weight updates.

Weight adjustment is performed using Gradient Descent or other modern optimization techniques.

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Chapter 3

Feature Selection

Feature selection is the process of selecting a smaller subset of features from the original feature set, with the goal of retaining the most essential information. This process helps reduce data dimensionality, improve model performance, and prevent overfitting. To select the optimal features, the Sequential Forward Floating Selection (SFFS) method was used in combination with 5-fold cross-validation (CV). [10]

SFFS belongs to the category of wrapper methods, as it relies on training a model to evaluate the features. The process starts with an empty feature set and incrementally adds the feature that offers the greatest improvement in model performance. Simultaneously, if a previously selected feature no longer contributes positively, it can be removed. This algorithm is classified as a greedy search algorithm, as it follows a locally optimal approach—selecting at each step the feature that provides the best immediate improvement, without considering all possible candidate solutions.

To evaluate the selected features, 5-fold cross-validation was applied. This technique splits the dataset into five equal-sized subsets. In each iteration, one subset is used as the test set, while the remaining four are used for training the model. This process is repeated five times so that each subset serves as the test set once.

The implementation was carried out using Python's mlxtend library.

Chapter 4

Statistical Analysis

Wilcoxon Signed-Rank Test

The Wilcoxon Signed-Rank Test (WSR) is a non-parametric ranking test used to compare two related samples. In this analysis, it was applied to determine whether the performance difference between the best classifier and the others is statistically significant. [11]

Test hypotheses in the present analysis:
- Null hypothesis (H0): There is no statistically significant difference in accuracy between the best classifier and the other classifiers.
- Alternative hypothesis (H1): There is a statistically significant difference in accuracy between the best classifier and the other classifiers.

Friedman Test

The Friedman Test is a non-parametric statistical test used to compare multiple (three or more) related groups. In this analysis, it was used to examine whether there are statistically significant differences in performance among all classifiers. [12]

Test hypotheses in the present analysis:
- Null hypothesis (H0): All classifiers have similar performance, and there are no statistically significant differences among them.
- Alternative hypothesis (H1): At least one classifier shows statistically significant performance differences compared to the others.

Mann-Whitney U Test

The Mann-Whitney U Test is a non-parametric test used to compare two independent groups. In this study, it was applied to investigate whether the selected features differ significantly between the two groups. This test is useful for feature analysis, as it allows the identification of the most discriminative features between groups. [13]

Test hypotheses in the present analysis:
- Null hypothesis (H0): The distribution of the feature values is similar in both groups, meaning the feature does not have discriminative power between the groups.
- Alternative hypothesis (H1): The distribution of the feature values differs significantly between the two groups, indicating that the feature has discriminative power and can be used for classification.

The SciPy library in Python was used to perform the above statistical tests.

Chapter 5

Hyperparameter Tuning for the Optimal Model

The selection of hyperparameters for the optimal model was performed using the GridSearchCV technique, combined with 3-fold cross-validation. This technique is widely used to evaluate all possible combinations of hyperparameters and identify the best possible configuration for the model.

The process begins by defining a grid of hyperparameters, which includes various values for each parameter to be optimized. GridSearchCV generates all possible combinations and trains the model on each of them.

For each hyperparameter combination, the model is trained and evaluated using 3-fold cross-validation. This process evaluates the model on three different data subsets to ensure the reliability of the performance. The average performance from the three subsets is used to estimate the model's overall performance.

Once all combinations have been trained and evaluated, GridSearchCV selects the combination that delivers the best performance, typically based on the accuracy metric.

Chapter 6

Model Evaluation Metrics

The performance of binary classification models is evaluated using various metrics. The most commonly used are Accuracy, Recall, Precision, F1 Score, and ROC-AUC. The values of these metrics range from 0 to 1, with values closer to 1 indicating better classifier performance. [14]

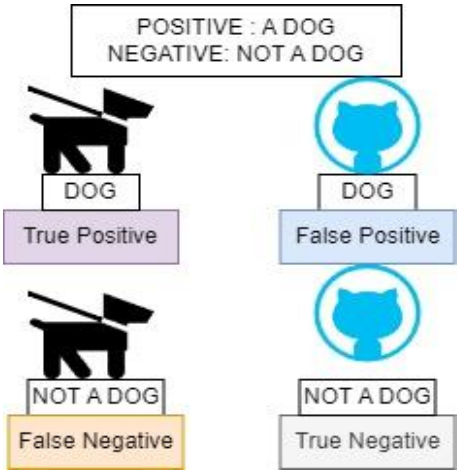
To compute these metrics, the data is categorized as follows:

True Positive (TP) – The classifier predicts positive and is correct.

False Positive (FP) – The classifier predicts positive but is incorrect.

True Negative (TN) – The classifier predicts negative and is correct.

False Negative (FN) – The classifier predicts negative but is incorrect.



Accuracy

Accuracy measures the ratio of correct predictions to the total number of predictions. It is calculated as:

$$\text{Accuracy}= \frac{TP+TN}{TP+TN+FP+FN}$$

Recall

Recall expresses how well the model identifies the positive cases. It is calculated as:

$$\text{Recall} = \frac{TP}{TP+FN}$$

Precision

Precision shows how many of the samples classified as positive are truly positive:

$$\text{Precision} = \frac{TP}{TP+FP}$$

F1-Score

The F1-score is the harmonic mean of Recall and Precision. It is especially useful for achieving a balance between the model's ability to detect positive samples and its ability to avoid false positive predictions.

$$F1 = 2 \times \frac{Precision \; x \; Recall}{Precision + Recall}$$

When the F1-score equals one, both precision and recall are also equal to one; otherwise, either recall or precision is equal to zero.

ROC-AUC

The ROC-AUC value measures a model's ability to correctly distinguish between two classes. The closer the ROC-AUC value is to 1, the better the model is at separating the data. The ROC-AUC is derived from the ROC curve, which is a graph that shows the relationship between Recall and the False Positive Rate (FPR) for different threshold values. The False Positive Rate (FPR) represents the proportion of negative samples that were incorrectly classified as positive.

Chapter 7

Literature Review

In 2021, a study was published that developed a machine learning model to predict the risk of mortality in COVID-19 patients up to 48 hours before death, using biochemical serum markers. The sample included a total of 398 patients, of whom 43 died. Out of the 26 features examined, the five most important ones—selected through logistic regression and coefficient analysis—were C-reactive protein, blood urea, serum calcium, serum albumin, and lactic acid. Based on these features, a Support Vector Machine (SVM) algorithm with a Radial Basis Function (RBF) kernel was trained. The trained SVM model with RBF kernel achieved 91% sensitivity and 91% specificity, with an AUC of 0.93 and an AUPRC of 0.76, indicating high classification ability and reliable performance. [15]

Another study investigated the potential of machine learning algorithms to predict mortality in hospitalized COVID-19 patients, evaluating several algorithms based on admission data from 1,500 patients (1,386 survivors and 144 deaths) at Ayatollah Taleghani Hospital in Abadan, Iran. A Genetic Algorithm (GA) was used to select the most significant prognostic factors, identifying dyspnea, ICU admission, and oxygen therapy as the primary features. Seven machine learning algorithms were then applied to build the mortality prediction model: J48 decision tree, Random Forest (RF), k-Nearest Neighbors (k-NN), Multi-layer Perceptron (MLP), Naïve Bayes (NB), eXtreme Gradient Boosting (XGBoost), and Logistic Regression (LR). Results showed that the Random Forest algorithm outperformed the others, achieving an accuracy of 95.03%, sensitivity of 90.70%, and specificity of 95.10%. [16]

A third study focused on predicting COVID-19 patient mortality using anonymized data from the UK Biobank, involving 14,877 patients, of whom 799 died. Feature selection was performed using the Regression Input Perturbation Ranking method in combination with the K-best algorithm based on the chi-square (χ^2) distribution. The Area Under the Curve (AUC) was used as the performance metric. Four machine learning models were applied in the analysis: Deep Neural Networks (DNN), Random Forest Classifier (RF), XGBoost Classifier (XGB), and Support Vector Machine (SVM). The Random Forest model demonstrated the best performance, with an AUC of 0.86 (95% CI: 0.84–0.88). The most significant prognostic factors identified were age, lifestyle, disease history, income, and family medical history. [17]

Chapter 8

Implementation

Dataset Description

In this study, anonymized data were used, referring to 73,000 individuals who were hospitalized with COVID-19 in Cyprus during the period 2020–2025. To manage the large volume of data and maintain balanced analysis, the downsampling technique was applied, resulting in a selection of 1,800 individuals, of whom 900 had died. The dataset includes 24 independent variables related to demographic, clinical, and hospital data of the patients, as well as one dependent variable indicating the outcome (death or recovery). Table 1 presents the variables used in the study along with their descriptions, while Table 2 includes the statistical characteristics of these variables.

Table 1
Variable – Description
- City: The city of origin of the individual
- Age: The individual's age
- Sex: The individual's gender
- Symptoms: Whether the individual exhibited symptoms
- Cough: Whether the individual had a cough
- Fever: Whether the individual had a fever
- SoreThroat: Whether the individual had a sore throat
- Myalgia: Whether the individual had muscle pain
- Diarrhea: Whether the individual had diarrhea
- RunningNose: Whether the individual had a runny nose
- Shortness_Breath: Whether the individual had shortness of breath
- Aosmia: Whether the individual had loss of smell
- Weakness: Whether the individual experienced weakness
- Shiver: Whether the individual experienced shivering
- Nausea: Whether the individual had nausea
- Notaste: Whether the individual experienced loss of taste
- Case_Classification: Describes how the person was infected with COVID-19, i.e., whether the infection occurred abroad, in Cyprus without known contact, through contact with a known case, or from someone who had traveled.
- Hospital: Whether the individual was hospitalized
- Morbitity: Whether the individual had any underlying condition (Diabetes, Cancer, Hypertension, Heart Disease, Kidney Disease, Immunosuppression)
- Icu_maf: Whether the individual was admitted to the ICU
- Ventilation: Whether the individual received mechanical ventilation
- Intubation: Whether the individual was intubated
- Vaccination: Whether the individual was vaccinated
- Reinfection: Whether the individual had a previous COVID-19 infection

Table 2

Variable – Categories – Frequency

- City:
    - Abroad: 5
    - Ammochostos: 103
    - British Bases: 1
    - Larnaka: 289
    - Lefkosia: 656
    - Lemesos: 603
    - Pafos: 143
- Sex:
    - Male: 0
    - Female: 1
- Symptoms:
    - Asymptomatic: 610
    - Symptomatic: 1190
- Cough:
    - No: 1313
    - Yes: 487
- Fever:
    - No: 1371
    - Yes: 429
- SoreThroat:
    - No: 1502
    - Yes: 248
- Myalgia:
    - No: 1472
    - Yes: 328
- Diarrhea:
    - No: 1712
    - Yes: 88
- RunningNose:
    - No: 1558
    - Yes: 242
- Shortness_Breath:
    - No: 1612
    - Yes: 183
- Aosmia:
    - No: 1671
    - Yes: 129
- Weakness:
    - No: 1411
    - Yes: 389
- Shiver:
    - No: 1596
    - Yes: 204
- Nausea:
    - No: 1721
    - Yes: 79
- Notaste:

- No: 1661
- Yes: 139

- Case_Classification:
  - Imported case diagnosed after travel: 33
  - Locally infected case not linked to known exposure: 1123
  - Contact of a known case (Secondary): 639
  - Infected by someone who had traveled (Secondary and Imported): 5
- Hospital:
  - Not hospitalized: 934
  - Hospitalized: 866
- Morbitity:
  - No underlying condition: 1272
  - Has an underlying condition: 528
- Icu_maf:
  - Not admitted to ICU: 1542
  - Admitted to ICU: 258
- Ventilation:
  - Not ventilated: 1765
  - Ventilated: 35
- Intubation:
  - Not intubated: 1575
  - Intubated: 225
- Vaccination:
  - Not vaccinated: 1032
  - Vaccinated: 768
- Reinfection:
  - Not reinfected: 1750
  - Reinfected: 50
- Age:
  - Min: 0
  - Max: 100
  - Mean: 55.17
  - Std: 26.09

Results

Table 1 presents the ROC-AUC performance metric of the Sequential Forward Floating Selection (SFFS) algorithm in combination with 5-fold cross-validation (CV), as a function of the selected features for the classification models: Logistic Regression, Random Forest, SVM with linear, polynomial, and RBF kernels, as well as MLP.

The summary of the average performance of the SFFS algorithm with 5-fold cross-validation for each machine learning model, by number of selected features, is presented in Figure 1 and Table 2.

Table 1

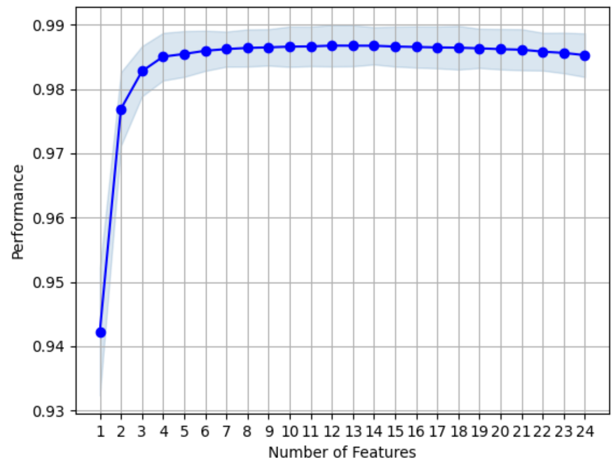
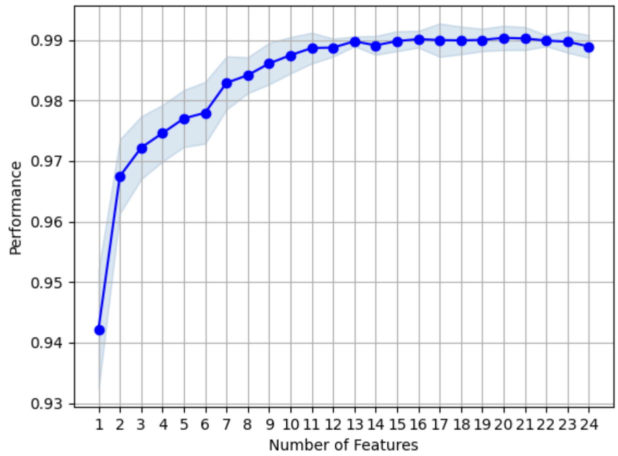
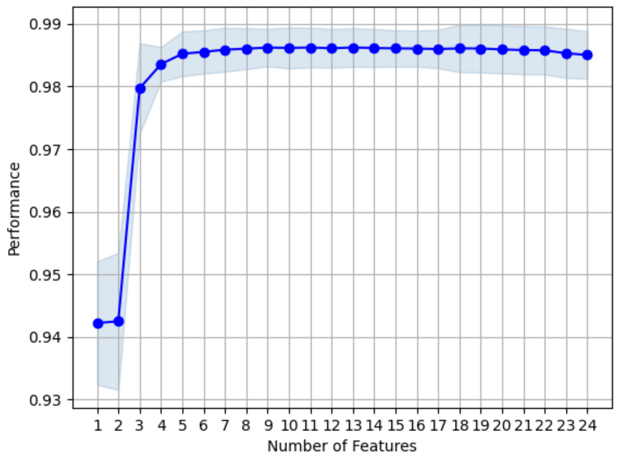
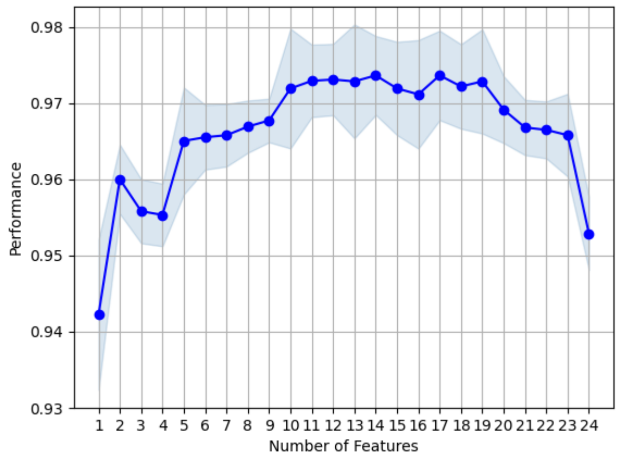
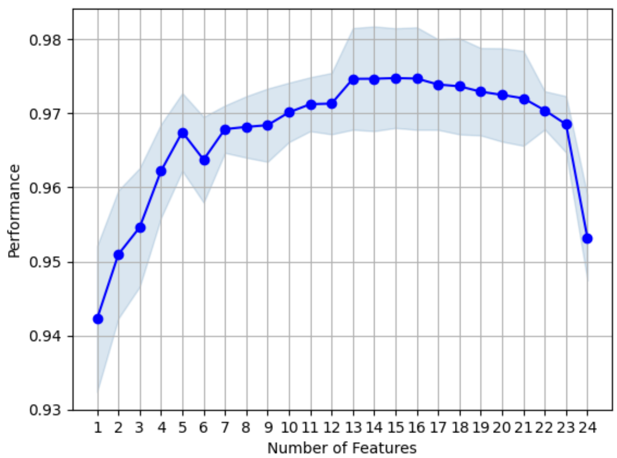
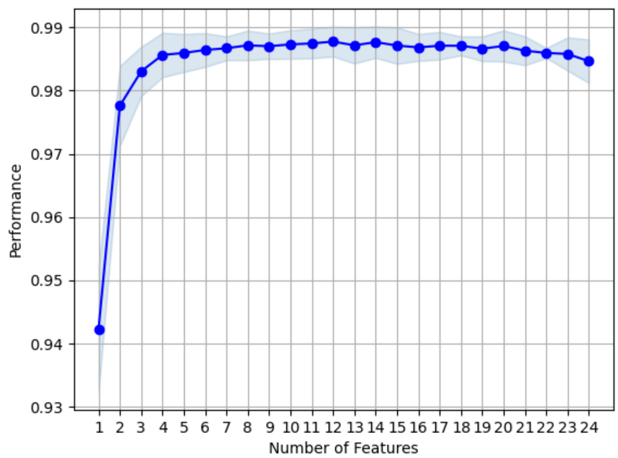
<table>
<tr><th>Logistic Regression</th><th>Random Forest</th></tr>
<tr><td>
</td><td>
</td></tr>
<tr><th>SVM_linear</th><th>SVM_poly</th></tr>
<tr><td>
</td><td>
</td></tr>
<tr><th>SVM_rbf</th><th>MLP</th></tr>
<tr><td>
</td><td>
</td></tr>
</table>

Figure 1

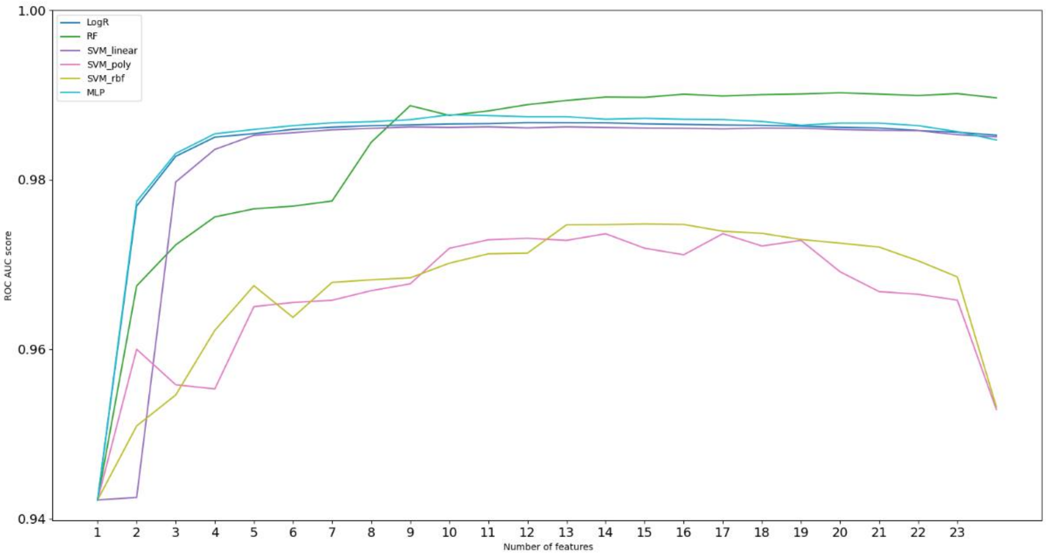


Table 2

Features	Logistic Regression	Random Forest	SVM_linear	SVM_poly	SVM_rbf	MLP
1	0.942222222	0.942222222	0.942222222	0.942222222	0.942222222	0.942222222
2	0.976888889	0.967469136	0.942512346	0.96	0.950950617	0.977611111
3	0.982753086	0.972194444	0.979725309	0.955808642	0.954604938	0.983049383
4	0.985012346	0.974598765	0.983567901	0.955320988	0.962222222	0.985641975
5	0.985435185	0.977018519	0.985231481	0.965027778	0.967493827	0.985962963
6	0.985932099	0.977962963	0.985537037	0.965515432	0.963753086	0.986435185
7	0.986191358	0.982895062	0.985882716	0.965774691	0.967882716	0.986703704
8	0.98637037	0.98417284	0.986052469	0.966904321	0.968185185	0.987166667
9	0.986459877	0.986098765	0.986222222	0.967712963	0.968419753	0.987027778
10	0.986570988	0.987469136	0.98616358	0.971904321	0.970148148	0.987314815
11	0.986611111	0.988651235	**0.986234568**	0.972910494	0.971256173	0.987466049
12	**0.986728395**	0.988731481	0.986111111	0.97308642	0.97133642	**0.987759259**
13	0.986716049	0.989765432	0.986231481	0.972839506	0.974679012	0.987169753
14	0.986709877	0.989101852	0.986157407	0.973623457	0.974697531	0.987675926
15	0.98658642	0.989796296	0.986092593	0.97191358	**0.974783951**	0.98712963
16	0.986518519	0.99012037	0.986061728	0.971141975	0.974719136	0.986842593
17	0.986450617	0.989962963	0.985987654	**0.97362963**	0.973916667	0.987108025
18	0.986407407	0.989916667	0.986092593	0.972175926	0.97367284	0.987095679
19	0.986308642	0.989996914	0.986074074	0.972839506	0.972944444	0.986638889
20	0.986179012	**0.99033642**	0.985919753	0.969135802	0.972512346	0.987070988
21	0.986098765	0.990228395	0.98582716	0.966796296	0.972049383	0.986317901
22	0.985796296	0.989882716	0.985783951	0.966478395	0.970429012	0.985966049
23	0.985598765	0.989694444	0.985308642	0.965787037	0.968533951	0.985799383
24	0.985243827	0.98891358	0.985046296	0.952885802	0.953194444	0.984657407
MEAN	**0.983824588**	**0.983633359**	**0.981918596**	**0.966309799**	**0.967275334**	**0.984326389**
STD	**0.00909442**	**0.011027732**	**0.012257577**	**0.007931672**	**0.008684486**	**0.009209273**

For the two best classifiers, Random Forest (ROC-AUC = 0.99033642) with 20 features and MLP (ROC-AUC = 0.987759259) with 12 features, the Wilcoxon signed-rank test was conducted for paired comparisons with the other classifiers. Table 3 presents the statistical values and p-values of the Wilcoxon test between these two classifiers and the remaining machine learning models. Table 4 includes the results of the Friedman test, showing the test statistic and corresponding p-value for comparing all classifiers.

Table 3

ML Classifiers	**Random Forest**	
	statistic	p-value
LG	156	0.292028
SVM_linear	183	0.08551
SVM_poly	276	1.35E-05
SVM_rbf	276	1.35E-05
MLP	145	0.415701

ML Classifiers	**MLP**	
	statistic	p-value
LG	263	7.18E-05
SVM_linear	274	1.76373E-05
SVM_poly	276	1.3508E-05
SVM_rbf	276	1.3508E-05
RF	131	0.584299

Table 4

	statistic	p-value
All ML classifiers	91.81987577639748	2.7854918549425427e-18

In Table 5, the MLP is selected as the optimal model, and the Mann-Whitney U test is applied to its selected features in order to evaluate the significance of these features and understand their impact on the model's final performance.

Table 5

Feature	U stat	p-value
Age	48151.5	7.757228387412299e-230
Symptoms	540900.0	4.3780651595137036e-51
Myalgia	487800.0	2.8540518521724423e-29
Diarrhea	418500.0	0.0010449345074944854
RunningNose	477900.0	4.5562245143428474e-29
Shortness_Breath	373050.0	3.096715336041267e-08

Weakness	467550.0,	1.752396864894406e-15
Hospital	46800.0	2.042114359130623e-308
Morbitity	325800.0	8.323662001169466e-20
Icu_maf	292500.0	2.0165669249493936e-63
Vaccination	494100.0	3.978609355073578e-21
Reinfection	402300.0	0.389697433408983

Table 6 presents the results with the hyperparameters selected for the optimal MLP model. Table 7 shows the performance metrics of the MLP model, including F1-score, Accuracy, ROC-AUC, Precision, and Recall.

Table 6

activation	alpha	hidden_layer_sizes	learning_rate	solver
tanh	0.0001	(20, 10, 10)	constant	adam

Table 7

	metric				
	accuracy	f1_score	roc_auc	precision	recall
score	0.956667	0.956425	0.956667	0.961798	0.951111

Conclusions

This study investigated the potential of machine learning algorithms for predicting COVID-19 mortality, with a particular focus on the selection of the most important features using the Sequential Forward Floating Selection (SFFS) method. The results indicated that the Multi-Layer Perceptron (MLP) and Random Forest models achieved the highest performance, with ROC-AUC scores of 0.9878 and 0.9903, respectively. However, the MLP required only 12 features, compared to 20 for Random Forest, making it a more efficient and interpretable model.

Statistical analysis using the Wilcoxon signed-rank test and the Friedman test confirmed that there were significant performance differences among the classifiers. Specifically, the MLP demonstrated statistically significant differences (p-value < 0.05) compared to Logistic Regression (LG), SVM_linear, SVM_poly, and SVM_rbf, while no significant difference was found when compared to Random Forest (p = 0.584299). Similarly, Random Forest showed significant differences with SVM_poly and SVM_rbf, but not with LG (p = 0.292028), SVM_linear (p = 0.08551), or MLP (p = 0.415701).

Feature analysis using the Mann-Whitney U test revealed that the feature "Reinfection" did not significantly contribute to the model’s performance and was subsequently removed from the dataset.

The performance metrics of the MLP model demonstrated its excellent predictive capability. The accuracy reached 95.67%, indicating that the model correctly classified the vast majority of cases, whether survivors or non-survivors. The recall was 95.11%, suggesting the model successfully identified nearly all actual non-survivors, reducing the risk of missing critical cases. At the same time, the precision (96.18%) indicated a very low rate of false positives, which is particularly important in clinical applications where incorrect predictions may lead to unnecessary or harmful interventions. The F1-score (95.64%) harmoniously combined precision and recall, confirming the balanced performance of the model. Although a slight reduction in ROC-AUC (0.9567) was observed compared to previous trials, the value remains high, indicating that the model effectively distinguishes between survivors and non-survivors.

In conclusion, this study highlighted the importance of feature selection and the use of advanced machine learning algorithms in predicting COVID-19 mortality. The findings suggest that the MLP is an efficient and interpretable model, requiring fewer features while maintaining high performance. The proposed methodology can also be applied to other medical prediction tasks, enhancing diagnosis and patient management.

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